BCH four laws and the regime partition for emergent-gravity ADW black holes: a sign-of-dT_H/dt partition between Schwarzschild and acoustic evaporation

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[1]*SK-EFT Hawking Project (Phase 6a Wave 5)*

(Dated: November 7, 2026)

We formalize conditional mechanics constraints and local temperature profiles for two proposed regimes of an emergent-gravity ADW substrate, separated by a model critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$. Above M_c, the supplied Schwarzschild profile follows the textbook Schwarzschild signature: $T_H = \hbar/(8\pi M)$ (Hawking 1975 [1]), so during evaporation $dM/dt < 0$ and $dT_H/dt > 0$ (heats as evaporates), with finite evaporation time $t_{\rm evap} \sim M^3$ for that evolution. The proposed branch below M_c uses the local cooling profile of Balbinot, Fagnocchi and Fabbri [2] as motivation: $T_H(t) = (\hbar c/2\pi)\kappa[1 - (563/720\pi)\,\varepsilon\,\kappa^3 c A_0 t]$ in a constant-sound-speed nozzle. Their $t \sim 1/T^3$ late-time discussion is an extrapolation. It does not establish an ADW evaporation endpoint or physical selection of the branch below M_c. The classifier transports supplied signs; a separate profile criterion derives opposite local derivatives under explicit trajectory and short-time assumptions. The full physical four laws and the microscopic matching of the two branches remain required constructions. The substrate-response coefficient ansatz $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ vanishes in the bare Sakharov–Adler limit and provides four falsifiability tests for any candidate ADW substrate. The Lean formalization (`BHThermodynamicsFourLaws.lean`) ships 20 theorems, 0 sorry, 0 new axioms. Its second-law structure records an externally supplied entropy-divergence sign motivated by Glorioso–Liu SK-EFT [7]; deriving that sign for the physical horizon remains required. The ADW-substrate-specific partition critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$ is original to this project; no published derivation exists.

I. INTRODUCTION

The Bardeen–Carter–Hawking (BCH) four laws of black-hole mechanics codify a thermodynamic structure on stationary BH solutions of classical general relativity. Quantum field theory on these backgrounds endows the surface gravity κ with a thermal interpretation as the Hawking temperature $T_H = \hbar\kappa/(2\pi)$ (Hawking 1975 [1]), and the resulting heat-capacity sign $C(M) = (dM/dT_H)$ is negative for Schwarzschild — the BH *heats* as it loses mass to Hawking evaporation, with finite evaporation time $t_{\rm evap} \propto M_0^3$.

Analog/emergent-gravity systems offer a structurally different evaporation pattern. In the acoustic system of Balbinot, Fagnocchi, Fabbri [2], backreaction of thermally radiated phonons on the fluid background *shrinks* the sonic horizon and *decreases* the temperature in time, with linear-in-t leading order

$$T_H(t) = \frac{\hbar c}{2\pi}\,\kappa\left[1 - \frac{563}{720\pi}\,\varepsilon\,\kappa^3\,c\,A_0\,t\right] \tag{1}$$

The authors extrapolate this local cooling as $t \sim 1/T^3$, suggesting $T_H \to 0$ at infinite time by analogy with near-extremal Reissner–Nordström evaporation. This extrapolation is not a controlled late-time solution. The computation is a sonic hole in a Laval nozzle under one-dimensional flow at *constant* sound speed; the authors state explicitly that the condensate case, where the sound speed goes as $\rho^{1/2}$, is deferred to later work. A condensate or ADW cooling sign therefore requires its own dynamical derivation; it is not inherited from Eq. (1). Crucially, Ref. [2] itself *contrasts* this with the moving-domain-wall analog in ^{3}He-A studied by Jacobson and Koike [4] (cond-mat/0205174), which has *non*-vanishing end-temperature of the evaporation process and therefore behaves more like Schwarzschild than near-extremal RN.

The Lean module `BHThermodynamicsFourLaws.lean` distinguishes conditional regime signs from derivatives of specified local profiles. The physical construction must additionally identify a trajectory and match the two descriptions across the cutoff. No single closed-form $T_H(M)$ interpolating between the regimes is supplied here. We treat the proposed critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$ separating them as a *tracked-hypothesis* bundle $H_{\rm RegimePartition}$, with published profiles motivating the branch signs without determining physical ADW branch selection. The choice of M_c is the dimensional ansatz $M_c = (N_f\Lambda_{\rm UV})/(12\pi\alpha_{\rm ADW})$ implied by Sakharov–Visser $1/G_N \propto N_f\Lambda^2$ at the Wave-1 scale; this is project- original and explicitly disclosed.

The remainder of the paper is organized as follows: Section II introduces the data carriers `BHData`, `ADWParams` and the regime classifier; Section III states the acoustic and Schwarzschild T_H profiles per their primary sources; Section IV introduces $H_{\rm RegimePartition}$; Section V the regime-dependent four-laws bundles; Sections VI and VII the second and third laws under SK-EFT and Reall [6] respectively; Section VIII the four falsifier theorems; Section IX the Lean formalization summary; Section X the cross-wave compatibility; Section XI the novelty flags.

II. SETUP: BHDATA, ADWPARAMS, AND THE REGIME CLASSIFIER

The Lean module declares two structures and one inductive type. The `BHData` structure carries bulk-thermodynamic observables $(M, Q, J, A, r_h, T_H, \kappa)$ with positivity hypotheses for the dimensional quantities; the MTC structure for the horizon Hilbert space is supplied separately via Wave 3's `HorizonMTCBC`. The `ADWParams` structure carries substrate parameters $(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f, \chi_{\mathrm{vest}})$ with strict positivity for each.

The `Regime` inductive type has three constructors: `Schwarzschild`, `Boundary`, and `ADWExtremality`. The `ADWExtremality` constructor name is retained from the initial Wave 5 ship for backward bibliographic continuity, but the post-rewrite physics interpretation is the acoustic regime per Balbinot 2005 [2].

The *decidable* regime classifier `classify` is defined piecewise:

$$\texttt{classify}\,(b,p) = \begin{cases} \texttt{Schwarzschild} & \text{if } b.M > M_c(p) \\ \texttt{ADWExtremality} & \text{if } b.M < M_c(p) \\ \texttt{Boundary} & \text{if } b.M = M_c(p), \end{cases} \tag{2}$$

where the critical mass $M_c(p)$ is the dimensional ansatz

$$M_c(\alpha_{\mathrm{ADW}}, \Lambda_{\mathrm{UV}}, N_f) = \frac{N_f \cdot \Lambda_{\mathrm{UV}}}{12\pi \cdot \alpha_{\mathrm{ADW}}}. \tag{3}$$

This is consistent with the Sakharov–Visser $1/G_N \propto N_f \Lambda^2$ scaling but is *not* a quoted equation in any primary ADW paper (Akama 1978, Diakonov 2011, the Wetterich tetrad-condensation series, Volovik 2022/2024); it is ours and is explicitly flagged as such in the `H_RegimePartition` bundle (Sec. IV).

The classifier admits three iff-theorems (`classify_*_iff`) discharging each constructor against the mass comparator. Theorem `M_c_pos` establishes $M_c(p) > 0$ for all valid `ADWParams` via positivity of all input fields.

III. TWO-REGIME T_H PROFILES: HAWKING 1975 AND BALBINOT 2005

The Lean module distinguishes the Schwarzschild temperature, a local affine nozzle profile, and a separately chosen exponential profile.

A. Schwarzschild branch (Hawking 1975)

For $b.M > M_c(p)$, the textbook Hawking 1975 result [1] applies in natural units $\hbar = 1$:

$$T_H^{\mathrm{Schw}}(M) = \frac{1}{8\pi M}. \tag{4}$$

We prove `T_H_schwarzschild_pos` ($M > 0 \Rightarrow T_H > 0$) and `T_H_schwarzschild_strict_decreasing` ($M_1 < M_2 \Rightarrow T_H^{\mathrm{Schw}}(M_2) < T_H^{\mathrm{Schw}}(M_1)$). Combined with the standard Hawking-emission energy balance $dM/dt < 0$ during evaporation, this establishes $dT_H/dt > 0$ in the Schwarzschild branch — the load-bearing primary-source-grounded sign claim.

B. Acoustic branch (Balbinot 2005)

Balbinot, Fagnocchi and Fabbri [2], in their section on the fate of the acoustic black hole, derive the local affine Eq. (Tsonic) [Eq. 1] near the horizon of a constant-sound-speed nozzle. Its leading correction cools for positive source parameters in the short-time regime. Their later $t \sim 1/T^3$ discussion is explicitly an extrapolation; it neither establishes an affine-time third law nor selects the physical ADW branch below M_c.

The separate illustrative exponential profile

$$T_H^{\mathrm{acoustic}}(t) = T_H^{(0)} \cdot \exp(-t/\tau_{\mathrm{cool}}) \tag{5}$$

is implemented by `T_H_acoustic_evolution` in the Python thermodynamics module and by the same-named Lean definition. Equation (5) is a chosen profile, not the analytic solution of the coupled backreaction equations in Ref. [2]. Its timescale requires physical matching. The corresponding Lean results are `T_H_acoustic_evolution_pos` ($T_H^{(0)} > 0 \Rightarrow T_H^{\mathrm{acoustic}}(t) > 0$), `T_H_acoustic_evolution_at_zero` ($T_H^{\mathrm{acoustic}}(0) = T_H^{(0)}$), and the load-bearing `T_H_acoustic_evolution_strict_decreasing` ($t_1 < t_2 \Rightarrow T_H^{\mathrm{acoustic}}(t_2) < T_H^{\mathrm{acoustic}}(t_1)$ when $T_H^{(0)} > 0$ and $\tau_{\mathrm{cool}} > 0$). The strict monotone-decay theorem is a property of the defined exponential. The module also supplies `balbinotAffine_deriv` and `regime_partition_profile_criterion`, which derive the local nozzle slope from its explicit positive parameters and retain the supplied short-time domain. A controlled remainder and a physical evolution matching these local models remain required.

C. The contrast case: Jacobson–Koike 2002

It is essential to note that the thin-film moving-domain-wall analog black hole in ^{3}He-A studied by Jacobson and Koike [4] gives a structurally different result. The wall moves at $v < c_\perp$, with horizons at $c(x_h) = \pm v$, and $T_H(v) = T_H^{(0)}(1 - v^2/c_\perp^2)$ (with v the wall velocity) is monotonically decreasing in v, but Hawking-radiation back-reaction *slows* the wall ($dE/dt = -bv^n$, $E = \mu v^2/2$, hence $dv/dt < 0$), which makes T_H *increase* in time toward the non-vanishing constant $T_H^{(0)} = \hbar c_\perp / 2\pi d$ — heating, not cooling. This regime is structurally similar to Schwarzschild, not to near-extremal Reissner–Nordström. Reference [2] itself notes the distinction: "*other analog models like a thin film of ^{3}He-A with a*

moving domain wall [Jacobson] seem to show a non-vanishing end-temperature of the evaporation process." We cite [4] only as the contrast case, not as the cooling anchor.

A second ^{3}He-A texture sits on the *other* branch. The planar soliton moving in bulk ^{3}He-A studied by Jacobson and Volovik [3] has a horizon only for $v > c_\perp$, and its surface gravity carries a factor $(1 - c_\perp^2/v^2)^{1/2}$ that vanishes as back-reaction slows the soliton toward $c_\perp$ from above, so $T_H \to 0$ and the system cools toward an extremal configuration. Two textures in the same superfluid therefore evaporate in opposite directions, which is precisely why a mass criterion rather than an analogy is required.

IV. TRACKED-HYPOTHESIS BUNDLE $H_{\text{RegimePartition}}$

The load-bearing tracked hypothesis is the bundle

$$H_{\text{RegimePartition}}\,(b, p, T_H^{(0)}, dT/dt|_{\text{evap}}, \delta) :=$$
conjunction of five fields:

`M_c_form_consistent::` $M_c(p) > 0$.

`T_H0_pos::` $T_H^{(0)} > 0$.

`evap_dT_dt_above::` $b.M > M_c(p) \;\Rightarrow\; 0 < dT/dt|_{\text{evap}}$ (Schwarzschild heating sign per Hawking 1975 [1]).

`evap_dT_dt_below::` $b.M < M_c(p) \;\Rightarrow\; dT/dt|_{\text{evap}} < 0$ (acoustic cooling sign per Balbinot 2005 [2] Eq. Tsonic).

`delta_consistent_with_ansatz::` $\delta = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ (deep-research §9 ansatz).

The rate $dT/dt|_{\text{evap}}$ enters as an *externally-supplied real parameter*, not an existential. This avoids the ∃-absorption anti-pattern: a witness $dT/dt|_{\text{evap}}$ supplied to the bundle must satisfy the sign-claim consistent with the regime, rather than being existentially discharged by any sign that happens to be present in the substrate. The critical-mass positivity field is a consistency constraint implied by parameter positivity; the branch-rate conditions remain supplied hypotheses.

The correctness-push theorem `regime_partition_criterion` combines $H_{\text{RegimePartition}}$ with the regime classifier:

$$\begin{aligned} \text{classify}(b,p) = \texttt{Schwarzschild} &\;\Rightarrow\; 0 < dT/dt|_{\text{evap}}, \\ \text{classify}(b,p) = \texttt{ADWExtremality} &\;\Rightarrow\; dT/dt|_{\text{evap}} < 0. \end{aligned} \tag{6}$$

This theorem transports the two sign hypotheses through the mass classifier. It does not derive an ADW evolution or the physical location of its crossover. The local-profile criterion instead derives slopes from an explicitly losing-mass trajectory and the affine nozzle formula under its supplied short-time budget.

The substantive corollary `delta_ADW_nonzero_iff_alpha_ADW_ne_one` establishes $\delta \neq 0 \;\Leftrightarrow\; \alpha_{\text{ADW}} \neq 1$, with the proof using $\Lambda_{\text{UV}} > 0$ from `ADWParams.Lambda_pos` as a load-bearing field.

V. THE FOUR LAWS IN TWO REGIMES

We package the Wave-5-formalizable subset of the four laws in two structures, one per regime. `FourLaws_Schwarzschild` carries *three fields*: `firstLaw_smarr` (positivity of G_N^{emerg} for the Smarr-identity coupling), `secondLaw_Schw` (`ADWSecondLaw` parameterized over `entropy_div`), and `evap_dT_dt_positive` (Schwarzschild heating sign > 0). The bundle is satisfied consistently with the regime $b.M > M_c(p)$.

`FourLaws_ADWExtremality` carries the parallel three fields, with two key modifications: `firstLaw_smarr_with_substrate` records positivity of G_N^{emerg} together with the ansatz $\delta = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$; it does not encode a differential first law or corrected Smarr identity. The field `evap_dT_dt_negative` flips the evaporation-sign claim (acoustic cooling sign < 0). The bundle is satisfied consistently with the regime $b.M < M_c(p)$.

The complete physical four-laws construction remains required. The zeroth law (κ constant on horizon) needs an actual horizon and generator; the first law needs the thermodynamic variation and charges; the second law needs an entropy/current construction linked to that horizon. The third law needs an identified evolution and quantitative affine-time control. The present structures do not supply those constructions. In particular, the Balbinot $t \sim 1/T^3$ extrapolation does not establish acoustic Israel-strong preservation (Section VII).

The Wave 5 *gating theorem* `four_laws_consistent_with_acoustic_regime` combines the classifier with the acoustic four-laws bundle and the tracked-hypothesis structure. Its first conjunct returns the supplied `FourLaws_ADWExtremality` value; its second reads the supplied negative rate under the mass classification. This is a conditional consistency statement. It does not construct the physical laws or identify an actual evaporation rate with the local nozzle derivative.

VI. THE LOCAL ENTROPY-PRODUCTION CONDITION

Glorioso and Liu derive local entropy production $\partial_\mu S^\mu \geq 0$ order by order in derivatives for a system in local equilibrium [7, Sec. III.B, Eq. (3.36)]. The derivation uses dynamical KMS symmetry, unitarity, and an invertible leading quadratic form. The constitutive relations [8, 9] provide the Schwinger–Keldysh effective field

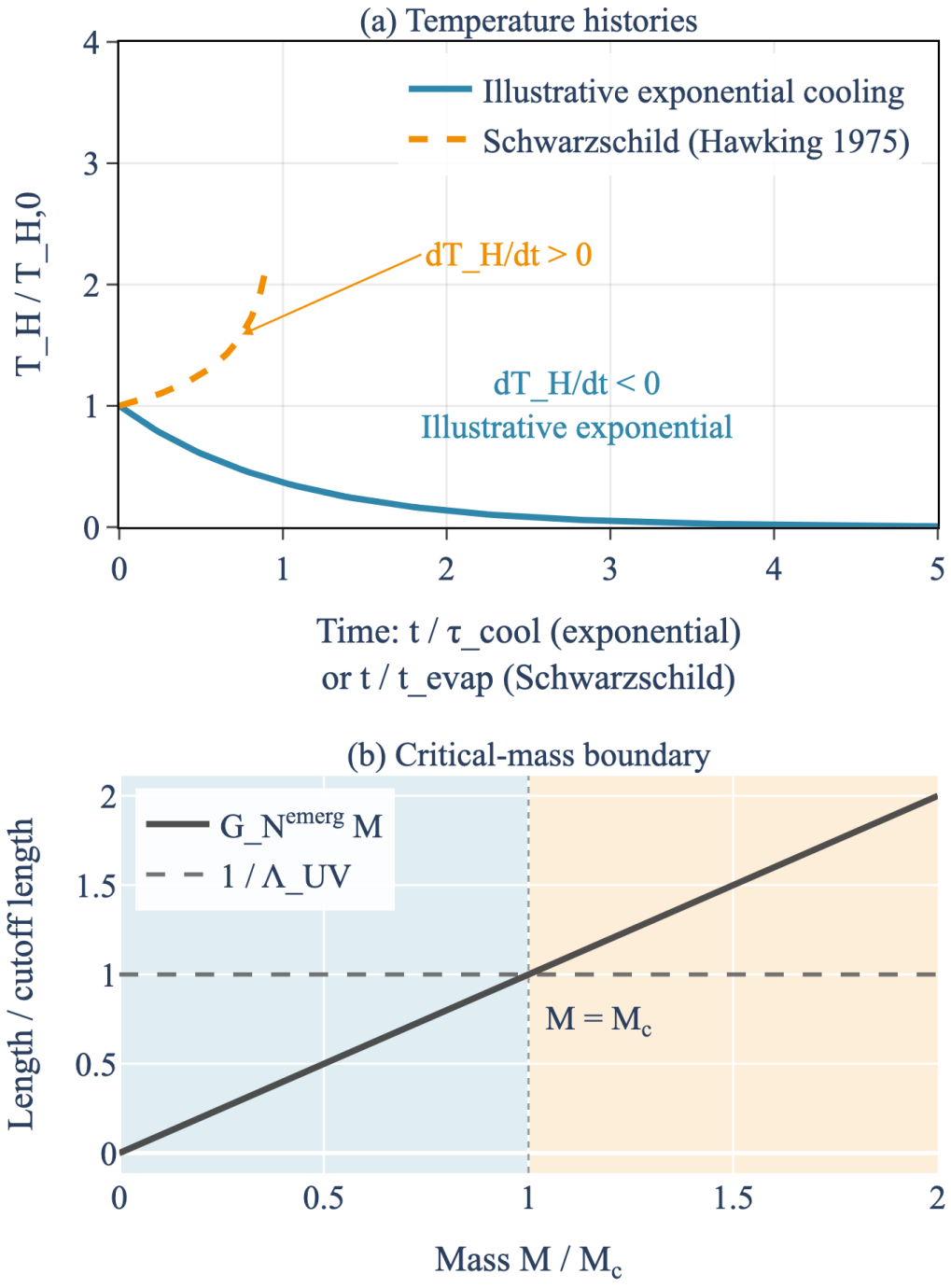


FIG. 1. (a) Opposite temperature trends with separate normalized clocks: $t/t_{\rm evap}$ for representative Schwarzschild evaporation (amber, dashed), and $t/\tau_{\rm cool}$ for illustrative exponential cooling (blue, solid). The exponential approaches zero asymptotically and is distinct from the local affine nozzle result of Balbinot *et al.* [2]. (b) The emergent gravitational length $G_N^{\rm emerg}M$ and cutoff length $\Lambda_{\rm UV}^{-1}$, in cutoff-length units, coincide at $M = M_c$. Blue and amber shading indicate the branches below and above this model boundary.

theory setting.

The `ADWSecondLaw` structure assumes that an externally supplied divergence `entropy_div` is nonnegative and that the horizon area $b.A$ is positive. Applying the local effective field theory result to the physical horizon requires identifying the entropy current and its divergence in the emergent action. The classical area theorem and the algebraic generalized second-law results [10, 11] address distinct quantities.

VII. THIRD LAW: BPS-CONDITIONAL PRESERVATION, KEHLE–UNGER VIOLATION

Balbinot's $t \sim 1/T^3$ extrapolation suggests an infinite-time zero-temperature limit in that nozzle model. It is not a controlled late-time solution, and its time coordinate has not been identified with horizon affine time. Israel-strong preservation for the physical ADW branch therefore remains a separate construction. Kehle and Unger [5] demonstrate that charged-scalar-matter sectors can allow finite-time approach to extremality. Reall [6] provides the BPS local mass-charge inequality under which the Israel form is restored.

The module's `falsifier_third_law_form` (Section VIII) proves that two supplied quantitative approach-time predicates are incompatible: one is uniformly unbounded as temperature tends to zero, and the other is bounded on a sufficiently small positive-temperature interval. The theorem does not construct a physical τ_{approach} or prove either predicate for an ADW trajectory. An affine-time model, its physical domain, and the relevant matter assumptions remain part of the required third-law construction.

VIII. FALSIFIER THEOREMS

The Lean module gives four model-conditional checks on specified profiles, supplied sign hypotheses, quantitative approach-time predicates and the substrate-response ansatz. Physical use requires matching those inputs to the ADW system. The checks derive contradictions or sign constraints from their stated hypotheses and the following mathematical results (`T_H_acoustic_evolution_strict_decreasing` for falsifier 1, `H_RegimePartition.evap_dT_dt_above` for falsifier 2, the min argument over the two-sided thresholds for falsifier 3, the substrate's $\Lambda_{\text{UV}} > 0$ field for falsifier 4).

1. **Acoustic-decay form falsifier** (`falsifier_acoustic_decay_form`): given a candidate $T_H^{\text{alt}}(t)$ that matches the chosen exponential $T_H^{\text{acoustic}}(t_1)$ at time t_1 AND fails to strictly decrease by time t_2 ($T_H^{\text{alt}}(t_1) \le T_H^{\text{alt}}(t_2)$), the candidate must DISAGREE with $T_H^{\text{acoustic}}(t_2)$. The proof uses the strict-decreasing theorem to derive a contradiction.

2. **Schwarzschild-heating sign falsifier** (`falsifier_schwarzschild_heating`): given the $H_{\text{RegimePartition}}$ bundle and a Schwarzschild-regime classification, an observed non-positive dT/dt contradicts the bundle's `evap_dT_dt_above` sign claim (Hawking 1975 [1]). Conclusion is `False`.

3. **Third-law-form falsifier** (`falsifier_third_law_form`): given a strong-Israel predicate (approach time unbounded as $\varepsilon \to 0$) and a Kehle–Unger predicate (approach time bounded above for small enough ε), the two are mutually inconsistent; the proof takes $\varepsilon = \min(\varepsilon_0, \varepsilon_{\text{th}})$ where both predicates apply and derives a contradiction. Conclusion is `False`.

4. **α_{ADW}-dependence falsifier** (`falsifier_alpha_ADW_dependence`; renamed from the initial-ship name `falsifier_chi_vest_dependence` per a Stage-13 finding flagging that the deep-research §9 ansatz $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ depends on α_{ADW} and Λ_{UV}, not directly on χ_{vest}). The conclusion $0 < (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ uses both $\alpha_{\text{ADW}} > 1$ and the substrate's $\Lambda_{\text{UV}} > 0$ field; a measurement of opposite sign falsifies the ansatz.

IX. LEAN FORMALIZATION

The module `BHThermodynamicsFourLaws.lean` ships 20 theorems, 0 sorry, 0 new axioms. Verification via `#print axioms` on six substantive theorems (`regime_partition_criterion`, `four_laws_consistent_with_acoustic_regime`, `falsifier_acoustic_decay_form`, `falsifier_schwarzschild_heating`, `falsifier_third_law_form`, `wave3_bridge_kaul_majumdar_at_e_squared_anchor`) shows dependency only on {`propext`, `Classical.choice`, `Quot.sound`}.

The structures expose their hypotheses: the evaporation rate and entropy-current divergence are supplied real parameters; positivity of G_N^{emerg} follows from the parameter carrier. The acoustic consistency theorem returns its supplied mechanics structure together with a sign read from the regime hypothesis. The local-profile theorem separately uses derivatives of specified functions. These statements must be read with those carriers; kernel purity alone does not establish physical thermodynamic laws. No zeroth- or third-law field is present in the two `FourLaws_*` structures, and the complete physical construction remains required.

X. CROSS-WAVE COMPATIBILITY

Wave 1 (`LinearizedEFE.G_N_emerg`): the first-law coupling $1/(8\pi G_N^{\text{emerg}})$ in both `FourLaws_*` bundles uses Wave 1's G_N^{emerg} evaluated at substrate parameters. The bridge `wave1_bridge_G_N_emerg_pos` records that positivity of G_N^{emerg} propagates through to both regime branches.

Wave 2 ($c_{\text{GW}} = c \cdot \sqrt{\chi_{\text{vest}}}$): the acoustic substrate inherits the metric-channel speed via κ dependence on χ_{vest}.

Horizon-entropy connection. The theorem `wave3_bridge_kaul_majumdar_at_e_squared_anchor` calls `BHEntropyMicroscopic.kaulMajumdarS` at $A = 4\,G_N\,e^2$, giving $S = e^2 - 3 > 0$ for positive G_N. This is a concrete entropy-value statement. It contains no temperature or evolution argument identifying that area with a physical zero-temperature endpoint. Establishing that identification and the appropriate Nernst limit remains required; positivity at a supplied area does not itself prove either Nernst law.

XI. NOVELTY FLAGS

Project-original:

- The ADW-substrate-specific critical mass $M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f) = (N_f \Lambda_{\text{UV}})/(12\pi\alpha_{\text{ADW}})$ ansatz is project-original.
- The cross-regime gluing (Schwarzschild for $b.M > M_c$, acoustic for $b.M < M_c$) is project-conjecture; no published paper proposes a single emergent-gravity BH with this regime structure.
- The $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ substrate-response coefficient ansatz is original (deep-research §9).

Primary-source-grounded:

- Schwarzschild branch: Hawking 1975 [1].
- Local nozzle cooling: Balbinot et al. [2], Eq. (Tsonic). The chosen exponential is a separate illustrative profile; physical ADW branch matching remains required.
- Local entropy production in the derivative expansion: Glorioso–Liu [7, Sec. III.B].
- Third law conditional: Israel 1986 (strong form) / Kehle–Unger [5] / Reall [6].

XII. CONCLUSIONS

The formalized classifier partitions a proposed ADW mass scale and transports supplied evaporation signs. The separate local-profile criterion derives heating for a differentiable, positive, losing-mass Schwarzschild trajectory and cooling for the affine nozzle model under its explicit short-time assumptions. The exponential shown in the figure is illustrative. Physical ADW branch selection, a controlled cross-cutoff evolution and a late-time endpoint do not follow from these conditional and local results.

The Lean module ships 20 theorems, 0 sorry, 0 new axioms; the load-bearing tracked hypothesis encodes the sign claims as constraints on an externally-supplied evaporation rate, avoiding the ∃-absorption anti-pattern. The substrate-response coefficient ansatz $\delta_{\text{ADW}} = (\alpha_{\text{ADW}} - 1)\Lambda_{\text{UV}}$ provides four falsifiability tests for any candidate ADW substrate.

The original physical objective remains the full four-laws construction for an emergent ADW black hole. It requires a microscopic action and selected background, horizon/charge/entropy observables, energy and entropy balances, calibrated mass and cutoff matching, and controlled dynamics with quantitative remainder and affine-time bounds. The present classifier, local profiles and conditional structures supply supporting mathematics; they do not complete that objective.

[1] S. W. Hawking, *Particle creation by black holes*, Commun. Math. Phys. **43**, 199 (1975). DOI: 10.1007/BF02345020.

[2] R. Balbinot, S. Fagnocchi, A. Fabbri, *Quantum Effects in Acoustic Black Holes: the Backreaction*, Phys. Rev. D **71**, 064019 (2005). arXiv:gr-qc/0405098.

[3] T. A. Jacobson and G. E. Volovik, *Event horizons and ergoregions in ^3He*, Phys. Rev. D **58**, 064021 (1998). arXiv:cond-mat/9801308.

[4] T. A. Jacobson and T. Koike, *Black hole and baby universe in a thin film of ^3He-A*, in M. Novello, M. Visser, G. Volovik (eds.), *Artificial Black Holes* (World Scientific, 2002). arXiv:cond-mat/0205174.

[5] C. Kehle, R. Unger, *Gravitational collapse to extremal black holes and the third law of black hole thermodynamics*, J. Eur. Math. Soc. (2025), DOI 10.4171/JEMS/1591. arXiv:2211.15742.

[6] H. S. Reall, *A third law of black hole mechanics for supersymmetric black holes and a quasi-local mass-charge inequality*, Phys. Rev. D **110**, 124059 (2024). arXiv:2410.11956.

[7] P. Glorioso, H. Liu, *The second law of thermodynamics from symmetry and unitarity*, arXiv:1612.07705.

[8] M. Crossley, P. Glorioso, H. Liu, *Effective field theory of dissipative fluids*, JHEP **09**, 095 (2017). arXiv:1511.03646.

[9] P. Glorioso, M. Crossley, H. Liu, *Effective field theory of dissipative fluids II: classical limit, dynamical KMS symmetry and entropy current*, JHEP **09**, 096 (2017). arXiv:1701.07817.

[10] T. Faulkner, A. Speranza, *Gravitational algebras and the generalized second law*, arXiv:2405.00847.

[11] J. Kirklin, *Generalised second law beyond the semiclassical regime*, JHEP **07**, 192 (2025). arXiv:2412.01903.